

COMPUTER NETWORKING I

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Introduction

The Universidad Distrital Francisco José de Caldas requires a high performance network infrastructure, to grant access to every faculty and service. This project analyzes the costs of implementation of a integral network, having in sigh things like bandwidth, topology, security and scalability. Two possible budgets will be shown, one thinking on the performance, realiability and the needs of the network, the other one will be cheaper and optimized for efficiency and budget.

Goal

GENERAL OBJECTIVE

To analyze, design, and estimate the costs of a network infrastructure for the Distrital University Francisco Jose de Caldas, granting high performance, security and escalability. Two possibles budgets will be made.

SPECIFIC OBJECTIVES

Specific objectives for this project are:

- Define the functional and technical requirements of the network.
- Comparing two cost estimations, an ideal estimation and a cost-optimized estimation.
- Propose an implementation and management strategy for implementation and future maintainance.

Proposed solution

SERVICES

The following services has to be included in the digital ecosystem:

- A main institutional website (www.udistrital.edu.co)
- Around 50 faculty and administrative websites
- A Virtual Learning Environment (Aulas Virtuales)
- The Academic Management System (SGA)
- A digital repository for research and publications
- More than 20 additional online platforms and services.

POPULATION

Faculty	Stimated Population
Sciences and Education	7.028
Engineering	5.243
Faculty of Environmental	3795
Technology Faculty	3.795
Faculty of Arts ASAB	3.795
Faculty of Mathematics	3.795
TOTAL	27.447

USERS

User Group	Stimated Population	Average Devices per User	Stimated Connections
students	29.102	2 (Laptop and Smartphone)	52.000
Faculty and Staff	3.000	2 (Laptop and Smartphone)	6.000
Visitors	500-1000	1.5	3.000-6.000
IoT and institutional Devices	5.000	1	5.000
Total	37.102 - 39102	66.000 - 69.000 devices	66.000-69.000 conections

SERVICES

Service	Purpose	estimated AP's
Academic Management System (SGA)	Enrollment, schedules, grades	30.000+ students
Virtual Learning (Aulas Virtuales)	Online classes, assignments	20.000+ users
Institutional Email	Official communication	30.000+ active accounts
Library & Repositories	Digital books, research papers	10.000+ queries
Research & Computing Labs	Simulations, cloud computing	1000+ daily
Administrative & HR Systems	Payroll, hiring, internal docs	3.000+ users
Student Services	Well-being, financial aid, medical	10+ daily
Public Website	General info, faculty pages	1600+ daily

References

[1] Gabriela Martínez Eslava 20202020117, Omar Armando Neira Ordóñez 20192020110, and Edison Mauricio Riaño 20172020113. Computers networking i project, 2025. URL <https://github.com/OmarNeira/ComputersNetworkingIOmarNeira>.

Results

STORAGE

At 40% of peak concurrency, this estimation is for a whole year.

Service	Estimated Storage Usage (TB)
Academic Management System (SGA)	10 TB/year
Virtual Learning (Aulas Virtuales)	10 TB/year
Institutional Email	5 TB/year
Library & Repositories	10 TB/year
Research & Computing Labs	5 TB/year
Administrative & HR Systems	5 TB/year
Public Website	10 TB/year

Wifi Full Coverage

This estimation is with 70% of users connected at the same time.

Faculty	Total Users	Stimated Peak Users (70%)	Required Bandwidth (Gbps)
Sciences and Education	5.543	5.130	30 Gbps
Engineering	7.328	3.880	25 Gbps
Faculty of Environmental	4.136	2.895	20 Gbps
Technology Faculty	4.136	2.895	20 Gbps
Faculty of Arts ASAB	4.136	2.895	20 Gbps
Faculty of Mathematics	4.136	2.895	20 Gbps
TOTAL	33.775	23.895	150 Gbps

Wifi Full Coverage

Based on the wifi full coverage assuming 50-70 users per AP there will be around to 45 to 80 access points on each faculty as needed.

Faculty	Total Users	Stimated Peak Users (70%)	estimated AP's
Sciences and Education	7.028	5.130	80
Engineering	5.243	3.880	60
Faculty of Environmental	3795	2.895	45
Technology Faculty	3.795	2.895	45
Faculty of Arts ASAB	3.795	2.895	45
Faculty of Mathematics	3.795	2.895	45
TOTAL	27.447	23.895	320

Non Functional requirements

- 3-5 CRAC (Computer Room Air Conditioners) units depending on the heat load.
- Filtered air intake/exhaust to prevent dust and contaminants from entering.
- Precision Cooling Units for temperature-sensitive equipment.
- 5-10 smart humidity sensors across the facility.

Physical security

- 2-4 Fingerprint-secured doors for controlled entry.
- Contract with a third-party security company (24/7 monitoring).
- 12-20 high-resolution security smart cameras.
- Mantrap (Optional): Expensive but effective dual-door security system.
- Environmental Sensors for leaks, dust, humidity temperature.

Electrical safety protocol

- 3-5 high-capacity UPS (Uninterruptible Power Supply) units for redundancy.
- Electrical Circuit Breakers and Fuses across racks to prevent overloads.
- Organized Cable Management System design for safety and maintenance.

AWS university's web applications

